

一、

1. c (光速不变原理)

2. $0.8c$ ($5\sqrt{1-\beta^2} = 3$)

3. $\frac{m}{s}$, $\frac{5m}{3s}$ ^{25 m / 9 SI} ($\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{5}{3}$)

4. $\frac{1}{4}m_0c^2$; $\frac{\sqrt{3}}{2}c$

$$\begin{cases} E_0 = m_0c^2 \\ E = \frac{m_0}{\sqrt{1-\beta^2}}c^2 = \frac{5}{4}m_0c^2 \end{cases}$$

$$\left(\frac{m_0}{\sqrt{1-\beta^2}}c^2 - m_0c^2 = m_0c^2 \quad \sqrt{1-\beta^2} = \frac{1}{2}, \beta = \frac{\sqrt{3}}{2} \right)$$

5. $\frac{2m_0}{\sqrt{1-v^2/c^2}}$; 0

$$(E = 2 \cdot \frac{m_0c^2}{\sqrt{1-v^2/c^2}} = M_0c^2)$$

6. 0

7. $-\frac{q}{8\pi\epsilon_0 a}$

$$\begin{cases} 7. V_p = 0 & V_p - V_\infty = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{a} \\ & V_m - V_\infty = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{2a} \end{cases}$$

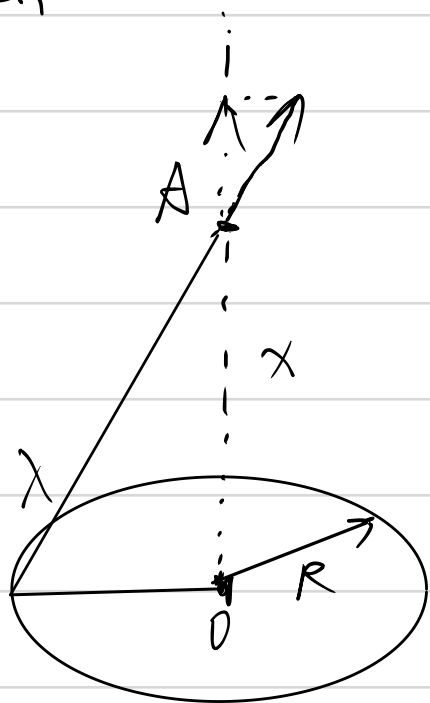
8. $(6-12xy)\vec{i} - 6x^2\vec{j} - 14z\vec{k}$

场强是电势的负梯度，应整体加负号

$$\begin{cases} U = 6x - 6xy - 7z^2 \\ U_x = 6 - 12xy, U_y = -6x, U_z = -14z \end{cases}$$

二、

1.



1) 显然在平行于圆环平面的方向 E 的分量为 0

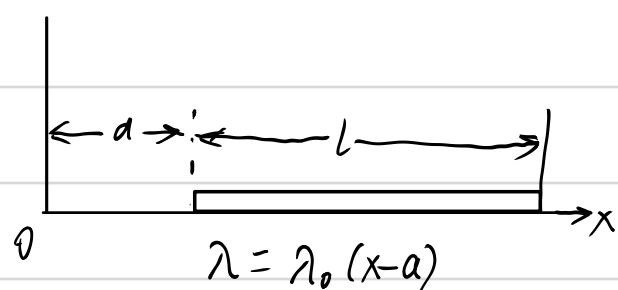
$$E = \int \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{R^2+x^2} \cdot \frac{x}{\sqrt{R^2+x^2}} = \frac{2\pi R\lambda x}{4\pi\epsilon_0(R^2+x^2)^{3/2}} = \frac{R\lambda x}{2\epsilon_0(R^2+x^2)^{3/2}}$$

若 $\lambda > 0$, $\|\vec{E}\| = \frac{R\lambda x}{2\epsilon_0(R^2+x^2)^{3/2}}$, 方向与 $\vec{OA}$ 相同.

若 $\lambda < 0$, $\|\vec{E}\| = -\frac{R\lambda x}{2\epsilon_0(R^2+x^2)^{3/2}}$, 方向与 $\vec{AO}$ 相同

(2) $F = QE = \frac{R\lambda x Q}{2\epsilon_0(R^2+x^2)^{3/2}}$ (惭愧, 未写方向讨论)

2.

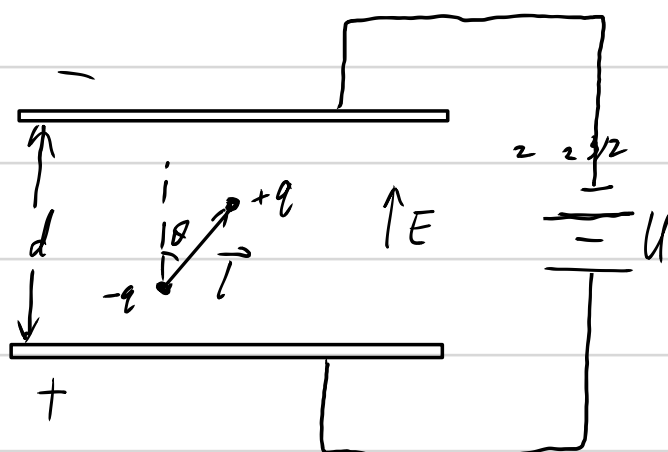


$$dV = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dx}{x} = \frac{\lambda_0}{4\pi\epsilon_0} \cdot \left(1 - \frac{a}{x}\right) dx$$

$$V = \int dV = \frac{\lambda_0}{4\pi\epsilon_0} \int_a^{a+l} \left(1 - \frac{a}{x}\right) dx$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left((l-a) \ln \frac{a+l}{a} \right)$$

3.



$$\|\vec{E}\| = \frac{U}{d}, \text{ 方向垂直向上}$$

$$\|\vec{F}_+\| = \frac{qU}{d}, \text{ 方向与 } \vec{E} \text{ 相同,}$$

$$\|\vec{F}_-\| = \frac{qU}{d}, \text{ 方向与 } \vec{E} \text{ 相反,}$$

受静电力合力为0

$$\|\vec{M}\| = \|\vec{F}_+\| \cdot \frac{1}{2}l \sin\theta + \|\vec{F}_-\| \cdot \frac{1}{2}l \sin\theta$$

$$= \frac{qUl \sin\theta}{d}$$

取 -q 所在处为电势能“0”点,

$$-qUl \cos\theta / d$$

$$\text{则 } E_p = -q\|\vec{E}\|l \cos\theta = -\frac{q^2Ul \cos\theta}{d}$$

$$4. \quad \rho = \begin{cases} \frac{qr}{\pi R^4} & (r \leq R) \quad (q > 0) \\ 0 & (r > R) \end{cases}$$

$$(1) \quad Q = \int_0^R \frac{qr}{\pi R^4} \cdot 4\pi r^2 dr = \frac{4q}{R^4} \cdot \frac{1}{4} r^4 \Big|_0^R = q$$

球内, $r \leq R$ 时,

$$Q_r = q \cdot \frac{r^4}{R^4}, \quad E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} Q_r = \frac{qr^4}{\epsilon_0 R^4},$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{qr^2}{R^4}, \text{ 方向为沿球心至该点处向外,}$$

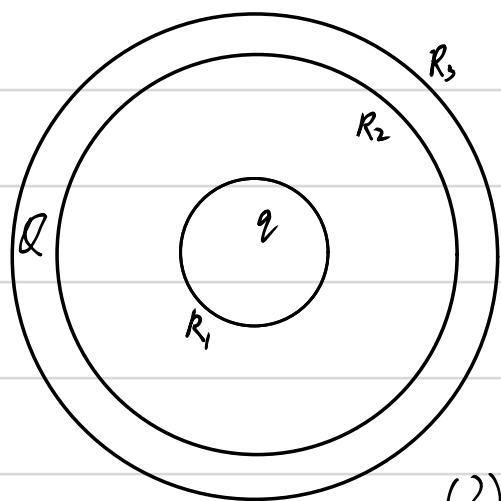
球外, $r > R$ 时, $E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} q, \quad E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}, \text{ 方向为沿球心至该点处向外.}$

$$\text{则 } \vec{E} = \begin{cases} \frac{1}{4\pi\epsilon_0} \cdot \frac{qr^2}{R^4} \vec{e}_r, & r \leq R \\ \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \vec{e}_r, & r > R \end{cases}$$

(3) 球外, $V = \int_r^{+\infty} \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$

球内, $V = V_R + \int_r^R \frac{1}{4\pi\epsilon_0} \frac{qr^2}{R^4} dr$
 $= \frac{1}{4\pi\epsilon_0} \frac{q}{R} + \frac{1}{12\pi\epsilon_0} q \cdot \left(\frac{1}{R} - \frac{r^3}{R^4} \right) = \frac{q}{3\pi\epsilon_0 R} - \frac{qr^3}{12\pi\epsilon_0 R^4}$

即 $V = \begin{cases} \frac{q}{3\pi\epsilon_0 R} - \frac{qr^3}{12\pi\epsilon_0 R^4}, & r \leq R \\ \frac{q}{4\pi\epsilon_0 r}, & r > R \end{cases}$



(1) $E = \begin{cases} 0, & r < R_1 \\ \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}, & R_1 < r < R_2 \\ 0, & R_2 < r < R_3 \\ \frac{1}{4\pi\epsilon_0} \cdot \frac{q+Q}{r^2}, & r > R_3 \end{cases}$

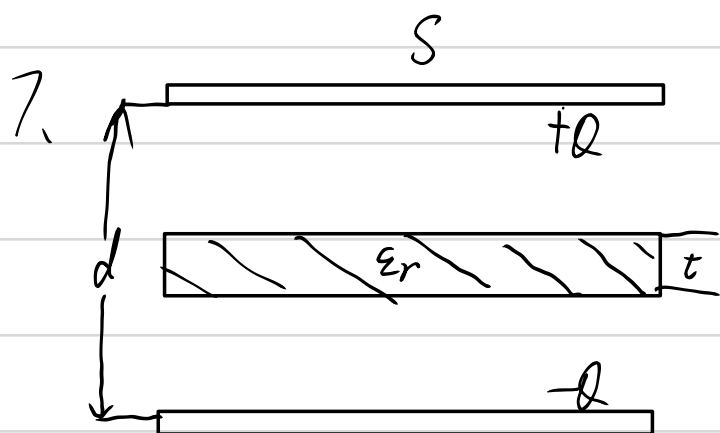
(2) 球 $V_1 = \int_{R_1}^{R_2} \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr + \int_{R_3}^{+\infty} \frac{1}{4\pi\epsilon_0} \frac{q+Q}{r^2} dr$
 $= \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R_1} - \frac{q}{R_2} + \frac{q+Q}{R_3} \right)$

$\frac{1}{R} V_2 = \int_{R_3}^{+\infty} \frac{1}{4\pi\epsilon_0} \frac{q+Q}{r^2} dr = \frac{1}{4\pi\epsilon_0} \frac{q+Q}{R_3}$

$\Delta V = V_1 - V_2 = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

6. (1) $W = \frac{1}{2} QV = \frac{1}{2} Q \cdot \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R} = \frac{Q^2}{8\pi\epsilon_0 R}$

(2) $m_0 c^2 = \frac{e^2}{8\pi\epsilon_0 R}, R = \frac{e^2}{8\pi\epsilon_0 m_0 c^2} = 1.41 \times 10^{-15} \text{ m}$



(1) $\sigma = \frac{Q}{S}$

$E_r = \frac{\sigma}{\epsilon_0 \epsilon_r} = \frac{Q}{\epsilon_0 \epsilon_r S}$

$D = \epsilon_0 \epsilon_r E_r = \frac{Q}{S}$

(2) $E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 S}$

$\Delta V = E(d-t) + E_r t$

$$\begin{aligned}
 &= \frac{Q(d-t)}{\epsilon_0 S} + \frac{Qt}{\epsilon_0 \epsilon_r S} \\
 13) \quad C = \frac{Q}{\Delta V} &= \frac{1}{\frac{d-t}{\epsilon_0 S} + \frac{t}{\epsilon_0 \epsilon_r S}} = \frac{\epsilon_0 \epsilon_r S}{\epsilon_r (d-t) + t}
 \end{aligned}$$

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